

Institute of Actuaries of India

Subject CP1 – Actuarial Practice (Paper A)

July 2022 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

Private equity investor. Preferable for values that represent an actuary's 'best estimate' of the future experience o but on aggressive end of the spectrum.

- Assumptions that are more likely to overstate/understate the liabilities or assets may lead to wrong decisions being made.
- But also have an interest in the price being as low as possible
- Management/Board of Directors. Will want the full picture to decide whether or not to proceed with the sale. o Therefore, they will want to see a range of figures on different assumptions.
- A stochastic approach could add value in this situation.
- Regulators (note two involved). May wish to see a realistic picture of the provider's finances or figures that intentionally understate the financial strength, o to ensure current policyholders continue to be protected.
- May also consider provisions on a discontinuance basis.
- Current shareholders. They will want to make their own assessment of the sale so will request to see an actuary's 'best estimate' value o but on the more prudent end of the spectrum
- But also want price as high as possible
- Policyholders. Will want to ensure security of benefits is not worsened post-transaction if provisions are undervalued
- DB scheme members. Similarly, will want to ensure security of their benefits is not worsened post-transaction
- Government/State. As it is a well-known company, the government may come under pressure to ensure that the company is not sold 'too cheap', leading to a loss of value, redundancies etc

[6 Marks]

Solution 2:

The framework will have the following key aspects:

- 1) List of asset to be included and what will be the metric to be used: (2 marks - The list needs to be exhaustive for candidate to get full marks).
 - a. GSec/STRIPS/Corporate - Duration
 - b. Equities – VaR, Impairment, Volatility index
 - c. MF- Duration/VaR
 - d. Derivatives – VaR, change in volatility
 - e. Real Estate – Vacancy risk, cashflow risk
 - f. FD/RDs – Comparison of rate of interest among banks
 - g. REITs, InvIT – Actual vs expected
 - h. Currency risk – movement in currency
 - i. Liquidity risk – Liquidity ratio over 3/6 or 9 months period
 - j. Counterparty risk (derivative)
 - k. ULIP fund performance vs benchmark - benchmark
- 2) What will be the KRIs to be monitored (2 marks for listing the key metrics as stated above)?
- 3) What will be the thresholds for the KRI as defined above?
- 4) What will be the frequency for collection of data and assessment?
- 5) Who are the risk owners for different KRIs that are needed?
- 6) Need to get the adequate data and plotting of numbers
- 7) RAG indicator calculation and assessment
- 8) Supporting commentary for RAG indicator
- 9) Commentary on strategic and tactical asset allocation.
- 10) Who are the audience for the dashboard resulting from the framework?

[7 Marks]

Solution 3:

i)

- High level preliminary risk analysis
- to confirm so high risk that it should not proceed.
- Hold a brainstorming session.
- With internal and external experts in the subject matter
- Carry out a desktop analysis.
- To supplement the brainstorming session identifying further risks and mitigations
- Construct risk register.
- Including cross reference to interdependencies

[4]

ii)

- Insufficient funds being set aside
- The underlying fund did not perform well and hence the fund not able to match the liability
- Insolvency of the provider o Due to investment underperformance
- Higher than expected expenses
- Or other reasonable example
- Partial withdrawal, if allowed, by the policyholder
- Investment mismatch
- Meaning full payment cannot be made – agreed with beneficiary
- Illiquidity of assets
- Fraudulent claim
- Death of beneficiary in the period
- Unexpected tax bill

[2]

iii)

- Benefits lower than expected.
- Cost of purchasing the intended item (eg annuity) has risen over the course of the period
- Inflation has eroded real return
- Leading to lower standard of living than expected

[2]

[8 Marks]**Solution 4:**

i)

Sources of surplus- Need a line or two describing how the profits will emerge

- 1) Mortality – experience profit resulting from actual experience being better than priced for.
- 2) Morbidity (only if Par policyholders have supported health business)
- 3) Claims amount – Actual vs Provisioned
- 4) Lapse /Surrender profit – release of reserves and surrender penalty
- 5) Investment income
- 6) One off investment gain on sale of assets
- 7) Expense management – resulting in lower acquisition and maintenance cost compared to what is priced for
- 8) Review of Asset Share or assumptions leading to release of reserve
- 9) Commission savings if higher than sales comes from direct channels than agency or bannca.
- 10) Taxation
- 11) Inflation impact being lower than expected
- 12) Premium -both new business and renewal. Higher than expected business and renewal would increase the profits.
- 13) One of profit transfer from Non par portfolio
- 14) Regulatory change leading to change in reserving leading to release of profits.
- 15) Better reinsurance arrangements resulting in lower reserves

[5]

ii)

1. Reduce the likelihood of claims through
 - 1.1. Good UW of new business
 - 1.2. Good UW at claims stage
 - 1.3. Providing customer incentives not to claim
2. Reduce the cost of claims though:
 - 2.1. Cost effective claims management procedures by reviewing ongoing claims
 - 2.2. Using reinsurance to limit the volatility of claims or protect from the risk of large claims
 - 2.3. Reducing future benefit payments
 - 2.4. Keeping guaranteed benefits to a minimum
3. Control expenses:
 - 3.1. Periodic review of expenses
 - 3.2. Keeping charges or premium flexible (if allowed)
 - 3.3. Ensuring claims experience are commensurate with the claim size
4. Using analytics to increase efficiency in claims management and investigation to reduce cost
5. Adopt an effective tax management policy

[5]

[10 Marks]

Solution 5:

i)

Describe algorithmic trading:

Algorithmic trading is a form of automated trading that involves buying or selling financial securities electronically to capitalise on price discrepancies for the same stock or asset in different markets. The parameters underlying the algorithm used to determine when assets would be bought or sold will need to be derived using data from an appropriate source(s).

Advantage:

- 1) Algorithmic tools can potentially lead to quicker, more consistent and fairer decisions being made. This has the potential to be good for consumers, businesses and society.
- 2) Advances in computing science mean that ever larger amounts of data can now be collected, stored and analysed much more quickly than in the past. These advances require new tools to analyse and make decisions.
- 3) Increasingly algorithmic tools are being used in decision making processes across many sectors. Advances in computer power, communication technology and programming capability have offered new tools for investment decisions, trading execution and risk management.
- 4) Electronic trading has the advantages of increased speed and efficiency of trading, and can result in lower dealing costs on trades. In addition, automated trading can potentially facilitate the execution of complex trading strategies that would not have previously been possible.

Disadvantages/risk:

- 1) There have been a number of examples where the use of algorithmic tools without appropriate care has created problems.
- 2) This has led to some concern from the public and regulators about the use of algorithmic decision making, particularly where the decisions could materially impact the lives of individuals.

- 3) It is important that steps are taken to ensure that risks and the potential for bias in decision making are managed.
- 4) Organisations will need to carefully consider what constitutes an appropriate level of due care in any algorithmic decision making. This will increase the cost of compliance and cost of doing business.
- 5) Possibility of regulator expecting a higher capital requirement to cover the downside risk.
- 6) There could be an error in the algorithm or the data used to parameterise the model could be wrong, leading to potential losses on each trade, rather than the expected profits. This is an issue when a large number of trades could be completed very quickly.
- 7) The algorithm may not operate properly in adverse conditions. For example, the algorithm could stop trading an asset in turbulent markets, reducing liquidity of the asset and increasing volatility.
- 8) In very turbulent conditions, trading in individual stocks, or even entire markets, may be suspended before an algorithmic trade can be completed.
- 9) The main risk of algorithmic trading is the possible impact on the financial system.

[6]

ii)

The answer needs to cover the key aspects like risk appetite, liability, understanding of algorithmic trading, cost of trading, disposable income, need for a phone etc.. Basis these factors the answer can be yes or a no but these factors are required as the basis.

[2]

iii)

Definition - A copula is a function, which takes as inputs marginal cumulative distribution functions and outputs a joint cumulative distribution function.

- 1) Copulas is a method used to aggregate risks to calculate capital requirement for a company as part of risk assessment and quantification.
- 2) They are another way of reflecting the dependence of the interrelated risks (like equity and bond portfolio) with each other, but rather being hidden within the joint distribution, it does so explicitly.
- 3) They are widely used in quantitative finance to model tail risk
- 4) It helps companies to minimise tail risks and helps optimize portfolios of investments.
- 5) It is used in capital assessment models to allow for aggregation of risks
- 6) By aggregating the risks effectively, it helps reduce the overall capital requirement, thereby optimizing the capital for the company.
- 7) Different forms of copula are available to suit different degrees of dependence

Risks:

- 1) Care needs to be taken that the management understands the complexity of the calculation before keeping capital basis it.
- 2) Choice of copula and its calibration can be difficult
- 3) Can be difficult to explain to the end user
- 4) There is always a possibility of manual error resulting in incorrect calculations.

[6]

[14 Marks]

Solution 6:

i)

- Liability Cover
- insure some of their high-value items of their choosing and also get discounts if their flatmates decide to join Building insurance (accidental damage) could also be provided.
- Cover away from the home can also be offered
- This could relate to taking possessions to and from student accommodation
- and also to more specialist cover. This could include cover for valuable items such as:

- Mobile phones, Laptops
- Car insurance
- Travel insurance, possibly including cover for gap year travel could be offered.
- Insurance could be provided for those studying abroad.
- Medical insurance, for countries where students are not covered by universal healthcare.
- Dental insurance
- Accident insurance
- Critical illness insurance
- Course fee insurance, providing cover for course fees due to cancellation or curtailment of the course.

[7]

ii)

- Some assumptions will be derived in the same way as for other lines of business – for example investment returns but many assumptions will need to be tailored for products offered under splitsurance.
- Historical data will usually be a primary source of data in determining assumptions about future experience, but the company may not have data relevant to the group of individuals(students) about whom assumptions are to be made.
- Past data will usually need to be adjusted in a subjective manner to allow for differences in the characteristics of the individuals concerned.
- Care will be needed over the choice of the assumptions that will have the most financial significance.
- With contents (high value items, as mentioned in the part i) insurance, the likelihood of theft may be higher as the accommodation is shared and it is unlikely that security measures are in place.
- The value of the contents is likely to be much lower than that of the company's usual business.
- For cover away from home, it is again likely that the value of the contents is different from that for the usual policies, for example in a shared house each resident may have their own television, laptop etc..
- For the more specialist insurance, the value of the sum insured will not be expected to be different from those of other such policies. Loss or damage may, however, be more likely. Whether and how assumptions are changed will depend on how much experience the company has with the particular specialist area.
- Students may have a different propensity to claim compared to typical policyholders, which may also lead to a different pricing assumption.
- Car insurance assumptions may be similar as they will already allow for age, occupation and NCD in their pricing. Will, however, need to ensure that the correct address is used when pricing these policies. Also, car may be used in sharing by the students who are of different age and hence the pricing may need to take account of that to refine the assumptions.
- Travel insurance will need different assumptions for the different types of insurance i.e. holiday, gap year, study abroad. Holiday insurance is likely to have similar assumptions to

standard travel insurance but could add cover for more adventurous activities to appeal to this market.

- The gap year and study abroad assumptions will be very different from that of their normal business and the country/countries to be visited will need to be taken into account.
- Medical insurance is likely to have similar assumptions to any other similar policies as age, sex and state of health will be allowed for.
- Course fee insurance is likely to be different from other insurance business written, prudence will be needed when setting assumptions for this business.
- Expense loadings will need to be adjusted as policies are likely to have lower premiums and lower sum assured as well as an increased likelihood of claims.
- Contingency loadings may be higher as this is a new market.
- Profit loading may depend on how competitive this market is.
- Recognising that students may stay with the same insurer after completing their studies, the insurer may price student insurance as a loss leader, hoping to make profit from retaining policyholders later in their careers.

[10]

[17 Marks]

Solution 7:

i)

- 1) Solvency is the key risk that the company will face.
- 2) Reasons for solvency being the risk:
 - a) Reduction in Asset Available for Solvency Margin (ASM) due to:
 - i) Significant impairment on the asset side resulting in significant drop in asset value
 - ii) Default on the expected coupons or maturity
 - iii) No new business being written thereby reducing the cash inflows
 - iv) Not able to and/or customers not willing to pay renewal premium
 - v) Expenses and contractual outgo to be paid to employees, customers and vendor there by reducing available capital
 - vi) Increase in reserve/provisions allowance resulting in reduction in profits or expected losses
 - vii) No retained profits for the company to absorb the stress
 - viii) Any additional taxes being imposed by the government thereby reducing the profits or increasing the outflow
 - ix) Significant claims arising out of protest leading to strain on financial position of the company.
 - b) Increasing in Required Solvency Margin (RSM)
 - c) Regulatory or internal pressure to review the assumptions upwards resulting in significant increase in solvency capital requirement.

[4]

ii)

1. **Financial Reinsurance or Fin Re** – Fin re is to exploit some form of regulatory arbitrage in order to manage the capital, solvency or tax position of a provider more efficiently. It frequently relies on the regulatory, solvency or tax position of a reinsurer who may be based in a overseas state, different from that of the provider/insurance company. This is done as a reinsurance contract between the reinsurer and the provider. One of the ways it is structured is that of a

“contingent loan” which while providing required financial support to the provider, does not show on their liability side of their balance sheet. It is mostly used to improve the solvency position of the company by crystallising the future expected profits from the in-force book of the provider.

2. **Securitization** - It involves an illiquid asset into a tradable instrument. The primary objective is to achieve regulatory or accounting “off balance sheet treatment”.
3. **Sub ordinate debt** – This is a form of raising debt from the capital market. Provider pays a regular interest/coupon rate on the debt to the buyers of the said debt and at maturity will pay back the principal amount. It is raised to improve the solvency of the company as it does not require to be reflected under liability while calculating solvency position.
4. **Banking products:**
 - 4.1. Liquidity facilities – Is used for short term financing for companies facing rapid business growth
 - 4.2. Contingent capital – Is a cost effective method of protecting the capital base of an insurance company. It is provided following deteriorating of experience (ie. That it is provided when needed.)
 - 4.3. Senior unsecured financing
5. **Derivatives** – Can be used to manage the liabilities so as to release capital. It can be used to either hedge (reduce risk and thereby capital) or speculate (take risk and increase the returns).
6. **Equity Capital** – raising equity capital will strengthen the solvency position of the company without impacting its liabilities. Equity may come from:
 - 6.1. From parent company
 - 6.2. From existing shareholders through rights issue
 - 6.3. Directly from market from new placement of shares.
7. **Internal sources of capital**
 - 7.1. Funds could be merged
 - 7.2. Assets could be merged
 - 7.3. The valuation basis can be weakened
 - 7.4. Capital could be retained in the company by not paying dividend to shareholders.

[7]

iii)

The key ways which may work for the company are below. Effectiveness of and likely use of any of the option will depend on regulatory and tax environment in the country.

- **Fin Re** - these are global reinsurers with global parent support and ability to provide loans. Company can trade its future profits from IF book in return of some capital. It is also a contingent loan, there is a possibility that regulator does allow not make any provisions for future payments on a statutory basis. This will help in strengthening cash position of the company.
- **Securitization** – It will work similar to Fin Re but the investors will be from capital market. Under current scenario, it may not be possible to issue any security to capital market either because there will be no investors given the uncertainty or required process couldn't be followed due to paucity of time.
- **Sub ordinate debt** – Similar to securitization, accessing capital market may not be possible or would be too costly for the company to raise the amount at reasonable cost.
- **Banking products:**
 - Liquidity facility
 - Contingent capital
 - Senior unsecured financing

All of the above would depend on state of banking sector under the current economic scenarios and their own capital position.
- **Loan from the government** or Central bank to stay afloat and maintain liquidity
- **Derivatives** - This is not an asset class which will help raise capital

- **Equity Capital** – Access to capital markets may not be possible or regulatory restrictions would be in place due to which this source of capital may not be possible.
- **Internal source of Capital** – This would be one of the methods that company will deploy under these circumstances, especially the revaluation of assets and liabilities, including deferring of all surplus distribution.

The most suitable option in the current scenario would be either:

- support from the regulator in easing solvency norm or
- loan from Central bank (under exceptional scenario).
- Review of the reserving assumptions resulting in reduction in capital requirements

[7]

[18 Marks]

Solution 8:

i)

- Reducing underwriting can also result in increased volumes of business sold. This is because distributors (and customers) may favour a more relaxed underwriting stance with less “hassle” factor (or intrusion) and reduced processing time.
- Higher volumes may lower costs thus leading to more attractive premiums (all else being equal) which may outweigh any increase in morbidity cost.
- Greater sales volumes could increase total profits to the company and reduce per policy expenses but may lead to admin/capital strain.
- There will also be the implementation costs for any new processes introduced,
- including any systems changes, amendments to the wording of application forms and proposals and retraining /redundancy costs. The company also needs to consider higher potential claims costs, the risk of increased anti selection and lapse and re-entry.
- If the company takes a very prudent approach given the reduced underwriting information, then more customers may be declined than previously. This could adversely impact the company’s reputation, which could have a knock on impact for sales volumes.
- Reducing claims underwriting is likely to lead to increased fraudulent claims being paid. In particular, there would be an increased risk of non-disclosure at the claims stage.
- Reduced financial underwriting may lead to moral hazard.
- The insurer will need to consider the level of underwriting carried out by its competitors for this product to ascertain whether it is currently generally in line with the market or is taking an overly cautious approach. A reinsurer may be able to offer advice on this.
- If reducing the level of underwriting would put the insurer out of line with its competitors then the insurer is likely to attract a disproportionate share of the adverse risks (i.e. anti-selection), which would increase the average morbidity experience of the portfolio.
- The insurer will want to charge for this additional morbidity risk and so is likely to have to increase premium rates. But this can further exacerbate the anti-selection effect, as the healthier lives are more likely to be take advantage of lower basic premium rates offered elsewhere where there is stricter underwriting.
- If there are any options available on the product (for example, increasing cover with no evidence of health), reducing initial medical underwriting will also increase the anti-selection risk on these options.
- Even if the impact of the lower underwriting costs exceeds the expected future additional claims costs, the company may not reduce its premium rates as it seems that the CFO is looking to reduce costs and increase profit margins rather than passing on all the cost reductions through to policyholders via lower premium rates.
- Further, if reinsurance is used then it is likely that the reinsurer will either increase its rates significantly if the level of underwriting is significantly reduced or in the extreme may not wish to continue reinsuring this product and this might also be reflected in higher premiums (or lower profit margin).

- Less underwriting will mean less homogeneity, making pricing harder. The lack of historic experience under the revised underwriting approach also makes parameter estimation and hence pricing more difficult. A margin might be included in premiums for uncertainty.
- If premium rates are increased then this is likely to be unpopular with the insurer's sales channels, which could offset the sales advantage from the reduction in the level of underwriting.
- The company may also need to hold higher reserves due to increased uncertainty about future morbidity experience. Increased reserves may affect the solvency of the company.
- There are other implications of lower sales (e.g. increased per policy overhead costs, lower market share) and these may not be implications that the CFO has intended

[10]

ii) .

- The company could reduce the number of cases that go through the underwriting process. For example, it could analyse how many cases are fully underwritten but where no additional premium is ultimately charged, to determine whether there are ways to avoid such cases unnecessarily going through the full underwriting process.
- The company could increase the Non-Medical limits at which further evidence is required e.g. increase the sum assured above which a medical test is required or increase the free cover limits if it writes group business. In considering this, it should compare the costs involved in obtaining further evidence at lower sums insured versus the additional premium charged and/or the savings in respect of cases refused.
- The company could include more detailed and objective questions on the application form.
- It could carry out less expensive on-line automated underwriting checks or use simplified telephone underwriting processes.
- The company could decline more cases rather than putting them through underwriting in order to determine an appropriate additional premium. For example, in the extreme, only those that answer all of the health questions positively could be offered cover.
- The company could rely more heavily on exclusion clauses or have simple rating scales for substandard risks. It could extend the waiting period or remove medical or lifestyle or financial underwriting.
- The company could improve the efficiency of its existing underwriting processes e.g. through introducing intelligent underwriting systems. The company could also review (and aim to increase) the number of cases handled by each member of underwriting staff and/or it could reduce the ratio of full underwriters to support staff.
- The company could improve staff training.
- The company could review and simplify processes.
- The company could use outsourcing, if cheaper.
- The company could reduce the fees paid to professionals or negotiate special rates or it could use lower qualified professionals, e.g. nurses rather than doctors for medical testing.
- If it is found that the underwriting of one particular product absorbs most of the underwriting resource, the insurer may focus on improving the efficiency for that particular product.
- The company could consider introducing a modified product that requires less underwriting. It could even stop selling that particular product.
- The costs will be higher for some of these proposals initially (implementation costs) but should be lower in the longer term.

[5]

iii) .

- The Company could introduce the wearables. Those who sync their devices with the Company's system and based on the data analytics, if they are found in good health, may be issued policies directly.
- This will not only reduce the cost of the company, but also reward the people in good health by fast issuance of policies and reduction in premium.

- The cost of device is there, but... people are willing to invest for health. A lot of people already have wearables,
- Insurers should be aware of the risks linked to reliance on new data sources, including the possibility that this data may prove to be unreliable. Safety recalls and privacy complaints also could negatively influence public perceptions.
- And all these challenges should be balanced against the significant opportunities these devices represent. The rewards could be great: data from wearable devices could help insurers better understand and adapt to changing customer needs and therefore increase placement ratios, improve customer engagement, and, perhaps more importantly, encourage active healthy living and improved claims experience.
- There are, of course, issues like who had been wearing them and who is taking the policy have to be cross- verified i.e. there may be moral hazard.
- When evaluating any novel data source, balancing the value of insight against the need to safeguard privacy is essential.
- Insurers would do well to consider four key factors when using data i.e. the four R's: regulations, reasonable expectations, risks (including data reliability), and rewards.
- By understanding the true value of wearable data, insurers may be able to more accurately underwrite, develop more attractive and affordable wellness initiatives that reduce the incidence and severity of chronic diseases, and improve the overall customer experience

[5]

[20 Marks]
